ABSTRACT

Intelligent Solid Waste Segregation Analysis is a sensor-based system developed to identify incorrectly disposed waste in biodegradable and non-biodegradable waste bins. In many places, people accidentally dispose of waste in the wrong bin, which makes waste collection and recycling difficult. This project aims to detect such mistakes and improve the accuracy of waste segregation.

The system uses an ESP32 microcontroller along with a gas sensor, moisture sensors, metal sensors, load cells with HX711 modules, level sensors, and LEDs. The moisture sensors help identify wet waste, while the metal sensors detect metallic materials. The gas sensor detects gases released near the waste. Load cells measure the weight of the waste, and level sensors monitor the filling level of the bins. The ESP32 processes the sensor readings to analyze the waste characteristics and identify possible incorrect disposal.

For example, when non-biodegradable waste such as plastic or metal is thrown into the biodegradable waste bin, the system checks the sensor readings to identify the unwanted material. Similarly, the system analyzes the waste in the non-biodegradable bin to identify possible biodegradable waste. LEDs provide visual indications when the sensor readings suggest incorrect waste disposal.

This project provides a simple and cost-effective approach to monitoring waste segregation and identifying disposal errors. It can be used in colleges, homes, offices, and public places to encourage proper waste disposal and improve waste management practices.

CHAPTER 1
INTRODUCTION

1.1 ABOUT THE PROJECT

Intelligent Solid Waste Segregation Analysis is a sensor-based system designed to identify incorrect waste disposal in biodegradable and non-biodegradable waste bins. In many places, people accidentally throw waste into the wrong bin, which affects waste collection, recycling, and disposal.

The proposed system uses an ESP32 microcontroller connected to gas, moisture, metal, load cell, and level sensors. These sensors collect information about the waste, including moisture content, the presence of metal, gas detection, weight, and bin filling level. The ESP32 processes the sensor readings to identify possible mismatches between the waste and the selected bin.

For example, if plastic or metal waste is thrown into the biodegradable waste bin, the system checks the sensor readings for signs of incorrect disposal. Similarly, it analyzes the waste in the non-biodegradable bin to identify possible biodegradable materials. LEDs provide visual indications based on the detected conditions.

The main purpose of this project is to reduce waste disposal errors and improve the monitoring of waste collection bins. The system can be used in educational institutions, residential areas, offices, and public places.

1.2 PROBLEM STATEMENT

Improper waste disposal is a common problem in waste management. Although separate bins are provided for biodegradable and non-biodegradable waste, people may accidentally dispose of waste in the wrong bin. This results in mixed waste, making recycling and further processing more difficult.

In conventional waste collection systems, the contents of the bins are generally checked manually. This requires additional effort and may not identify

incorrect disposal immediately.

Therefore, a sensor-based system is needed to analyze waste characteristics, identify possible incorrect disposal, and monitor the weight and filling level of the bins. The proposed system uses an ESP32 and multiple sensors to support waste analysis and provide visual indications when incorrect disposal is suspected.

1.3 OBJECTIVES

The main objectives of the project are:

1. To identify possible incorrect disposal of waste in biodegradable and non-biodegradable bins.
2. To use moisture sensors to detect the moisture content of waste.
3. To detect metallic materials using metal sensors.
4. To monitor gases released near the waste using a gas sensor.
5. To measure the weight of collected waste using load cells and HX711 modules.
6. To monitor the filling level of the bins using level sensors.
7. To process sensor readings using the ESP32 microcontroller.
8. To provide visual indications through LEDs when unwanted waste is suspected.
9. To reduce manual effort in monitoring waste bins.
10. To encourage proper waste disposal and improve waste management practices.

1.4 SCOPE AND MULTIDISCIPLINARY RELEVANCE

The project focuses on analyzing waste in biodegradable and non-biodegradable bins using sensors and a microcontroller. It aims to identify possible waste disposal errors and monitor bin weight and filling level.

The gas, moisture, and metal sensors provide information about the characteristics of the waste. Load cells measure the weight, while level sensors help determine how full the bins are. The ESP32 processes the sensor readings and controls the LED indications.

This project combines different areas of engineering. Electronics is used for sensor interfacing and circuit connections. Embedded systems are involved in programming the ESP32 and processing sensor data. Basic data analysis is used to interpret sensor readings and identify possible waste mismatches.

The system can be further developed by adding a display, wireless monitoring, data storage, or a mobile application. These improvements could help users and waste management personnel monitor bin conditions more effectively.

The proposed system is suitable for small-scale applications such as colleges, homes, offices, and public areas. Its effectiveness depends on the accuracy of the sensors and the rules used to interpret their readings.

CHAPTER 2
LITERATURE SURVEY

2.1 RELATED WORK (AI / DEEP LEARNING)

Artificial Intelligence and deep learning are used in some waste management systems to classify different types of waste. These methods generally require suitable datasets and trained models to identify waste categories.

However, the proposed project does not use artificial intelligence, deep learning, or image processing. It uses sensor readings to analyze waste characteristics and identify possible incorrect disposal in biodegradable and non-biodegradable bins. Therefore, the project focuses on sensor-based analysis rather than AI-based classification.

2.2 RELATED WORK (IoT / EMBEDDED / ELECTRICAL OR ELECTRONICS)

Sensor-based waste management systems use electronic components to monitor waste characteristics and bin conditions. Moisture sensors are used to measure the moisture content of waste, while metal sensors detect metallic materials. Gas sensors detect certain gases present near waste.

Load cells connected to HX711 modules measure the weight of collected waste. Level sensors monitor the filling level of the bins. A microcontroller such as the ESP32 processes the sensor readings and controls output devices such as LEDs.

The proposed project combines these components to analyze waste in biodegradable and non-biodegradable bins. The sensor readings are processed to identify possible incorrect disposal and provide visual indications when the programmed conditions are met.

2.3 LITERATURE GAP

In waste management, incorrect disposal can occur when biodegradable waste is placed in a non-biodegradable bin or vice versa. Manual inspection requires additional effort, while systems that only monitor bin weight or filling level do not provide sufficient information about waste characteristics.

The proposed project combines gas, moisture, metal, weight, and level measurements to analyze waste conditions. The ESP32 processes the sensor readings to identify possible mismatches between the waste and the bin category.

This project focuses on developing a simple, low-cost sensor-based system for detecting possible waste disposal errors and monitoring bin conditions. The system's performance depends on the sensor readings and the detection rules implemented in the ESP32.

CHAPTER 3
EXISTING SYSTEMS

3.1 METHODOLOGIES USED

Traditional solid waste management systems mainly depend on separate waste bins and manual waste collection. Biodegradable and non-biodegradable wastes are collected in different bins, and people are expected to dispose of waste according to its type.

In some systems, sensors are used to monitor the filling level of bins and determine when waste collection is required. Load cells can be used to measure the weight of collected waste, while level sensors help monitor the available space inside the bins.

Waste identification can also be supported by individual sensors. Moisture sensors provide information about the moisture content of waste, and metal sensors detect metallic materials. Gas sensors detect certain gases present near the waste.

However, these measurements may not be combined to identify incorrect disposal in conventional waste collection systems. Manual inspection may therefore be required to check whether the waste in a bin matches its intended category.

3.2 DISADVANTAGES

The disadvantages of conventional waste management systems are:

1. **Manual inspection:** Checking the contents of waste bins requires human effort.
2. **Incorrect waste disposal:** People may accidentally throw biodegradable

waste into non-biodegradable bins or vice versa.

3. **Mixed waste:** Incorrect disposal makes waste collection, recycling, and processing more difficult.
4. **Limited waste analysis:** Ordinary bins do not identify waste characteristics automatically.
5. **Difficulty in monitoring bin weight:** Systems without weighing equipment cannot provide direct weight measurements.
6. **Overflow problems:** Without bin-level monitoring, bins may become full before waste is collected.
7. **Delayed identification:** Incorrect disposal may remain unnoticed until the bin is inspected.
8. **Increased sorting effort:** Mixed waste may require additional manual separation before further processing.

These limitations show the need for a sensor-based system that analyzes waste characteristics and monitors bin conditions to help identify possible waste disposal errors.

CHAPTER 4
HARDWARE AND SOFTWARE DESCRIPTION

4.1 INTRODUCTION

The Intelligent Solid Waste Segregation Analysis system is developed using electronic sensors and an ESP32 microcontroller. The system is designed to analyze waste in biodegradable and non-biodegradable bins and identify possible incorrect disposal based on sensor readings.

The hardware consists of a gas sensor, moisture sensors, metal sensors, load cells, HX711 modules, level sensors, LEDs, a breadboard, jumper wires, and an ESP32 development board. Each component performs a specific function in collecting information about the waste and monitoring the condition of the bins.

The ESP32 acts as the central processing unit. It receives signals from the connected sensors and executes the programmed instructions. Based on the sensor readings and the conditions defined in the program, the system provides visual indications through LEDs.

The software is responsible for reading sensor values, processing the measurements, applying the programmed detection rules, and controlling the output indicators. The combination of hardware and software provides a practical approach to waste analysis and bin monitoring.

4.1.1 SOFTWARE REQUIREMENTS

The software requirements include the programming environment and the tools used to develop and upload the program to the ESP32.

ESP32 Programming Environment

The ESP32 requires a program to read sensor inputs and control the output devices. A compatible development environment can be used to write, compile, and upload the program to the microcontroller.

If Arduino IDE is used, the ESP32 board support package allows the

program to be compiled and uploaded to the development board. The program can be written using C/C++ syntax and appropriate libraries for the connected components.

Embedded Programming

Embedded programming is used to control the operation of the system. The program reads the sensor outputs, processes the measurements, and checks whether the readings satisfy the defined conditions.

The program can also control LEDs to indicate detected conditions. The exact detection rules depend on the sensor types, their calibration, and the logic implemented in the project.

Serial Monitor

A serial monitor can be used during development to display sensor readings and debugging messages. It helps verify whether the sensors are connected correctly and whether the ESP32 receives the expected input values.

The serial monitor is also useful when testing different waste samples and adjusting the thresholds used in the program.

4.2 HARDWARE REQUIREMENTS

The hardware components are selected to measure different characteristics of waste and monitor the condition of the collection bins.

The major components used in the proposed system are:

1. ESP32 microcontroller – 1
2. Gas sensor – 1
3. Moisture sensors – 2
4. Metal sensors – 2
5. Load cells – 2
6. HX711 modules – 2
7. Level sensors – 2
8. LEDs for indication

9. Breadboard – 1
10. Jumper wires – as required

The sensors provide input data, while the ESP32 processes the readings. The load cells measure weight through the HX711 modules, and the LEDs provide visual indications based on the program.

4.2.1 ESP32 MICROCONTROLLER

The ESP32 is a microcontroller widely used in embedded systems and Internet of Things applications. It provides digital input/output pins, analog input capabilities on supported pins, and built-in wireless communication features.

In this project, the ESP32 acts as the main controller. It receives data from the gas, moisture, metal, load-cell, and level-sensing units. It processes the inputs according to the program and controls the LEDs.

The ESP32 is suitable for this application because it supports multiple sensor connections and can execute the required monitoring logic. Its wireless capabilities also provide an opportunity for future improvements, such as remote monitoring and data reporting.

The number of available pins and the type of signal supported by each pin must be considered while connecting the sensors. The wiring should also ensure that the input voltage levels are compatible with the ESP32.

4.2.2 GAS SENSOR

The gas sensor is used to detect certain gases present near the waste. Organic waste may release gases during decomposition, while other materials and environmental conditions may also affect gas readings.

In the proposed system, the gas sensor provides an additional measurement that can be considered when analyzing the contents of the biodegradable and non-biodegradable bins.

The sensor output is read by the ESP32 and evaluated using the

programmed conditions. Depending on the sensor model, the output may be analog, digital, or both.

The gas sensor does not independently identify every type of waste. Its readings depend on the gas being detected, sensor sensitivity, distance from the waste, ventilation, and environmental conditions.

The exact gas sensor model and the gases it can detect should be specified in the final report.

4.2.3 MOISTURE SENSORS

Two moisture sensors are used to measure the moisture characteristics of waste in the two-bin arrangement.

Moisture content is an important characteristic in waste analysis. Food scraps and other wet organic materials often contain more moisture than many dry materials. However, moisture alone cannot reliably determine whether a material is biodegradable.

The moisture sensors provide readings to the ESP32, which evaluates them using the conditions defined in the program. The measurements can help identify waste that differs from the expected characteristics of the bin contents.

Using a separate sensor for each bin allows the two bins to be monitored independently, provided that the sensors are installed and connected accordingly.

The sensors should be tested with different waste samples to determine suitable operating thresholds. Their readings may vary depending on contact with the waste, sensor placement, and environmental conditions.

4.2.4 METAL SENSORS

Two metal sensors are included in the system to detect metallic materials. Metal detection is useful because metallic objects may be disposed of incorrectly in either biodegradable or non-biodegradable bins.

The metal sensors produce output signals when metallic materials are

detected within their sensing range. The ESP32 reads these signals and checks them against the programmed conditions.

A metal sensor can help identify the presence of metal, but it cannot determine the complete composition of a waste item. For example, an object made of plastic with a metallic part may also trigger detection, depending on the sensor and its placement.

Separate metal sensors can support independent monitoring of the two bins. Proper positioning and testing are necessary to reduce incorrect readings.

4.2.5 LOAD CELLS

Two load cells are used to measure the weight of waste. A load cell is a transducer that converts an applied force into a small electrical signal.

When waste is placed in a bin supported by a load cell, the applied load produces a change in the sensor's electrical output. This signal is then processed through an HX711 module.

The weight measurements help monitor the quantity of waste collected in each bin. They can also be used to observe changes in waste quantity over time.

Load cells require calibration before accurate weight measurements can be obtained. The calibration process involves comparing sensor readings with known weights and adjusting the conversion factor.

The load cells should be mounted securely so that the weight is transferred correctly to the sensing element. Their maximum rated capacity must not be exceeded.

4.2.6 HX711 MODULES

The HX711 is a precision 24-bit analog-to-digital converter designed for use with load cells and strain-gauge sensors. It amplifies the small electrical signal from a load cell and converts it into digital data that a microcontroller can read.

Two HX711 modules are used with the two load cells. Each module interfaces with its corresponding load cell and provides the measured signal to the ESP32.

The modules generally communicate with the microcontroller through two signal connections commonly labelled DT (data) and SCK (clock). The appropriate pins must be configured in the program.

The HX711 modules are important because the electrical signals generated by load cells are very small. Directly measuring these signals using ordinary microcontroller inputs would not provide the same level of precision.

Correct wiring, stable mounting, calibration, and suitable software libraries are necessary for reliable weight measurements.

4.2.7 LEVEL SENSORS

Two level sensors are used to monitor the filling condition of the biodegradable and non-biodegradable bins.

As waste accumulates, the available space inside a bin decreases. Level sensors provide information about the waste level and help identify when a bin approaches its capacity.

The ESP32 reads the sensor outputs and applies the programmed conditions to determine the indicated level. LEDs may be used to provide a visual indication when the bin reaches a predefined level.

The exact operation depends on the type of level sensor used. Some sensors provide digital outputs, while others provide analog measurements or operate using different sensing principles.

The sensors should be positioned to measure the intended filling level. Their output should be tested under different conditions to ensure that the readings are suitable for the selected bin design.

4.2.8 LEDs FOR INDICATION

LEDs are used as visual indicators in the proposed system. They provide a simple way to communicate the status of the system without requiring a separate display.

Depending on the implemented program, LEDs can indicate detected conditions, possible incorrect waste disposal, or a bin-level warning.

The ESP32 controls the LEDs through its output pins. Suitable current-limiting resistors should be used where required to protect the LEDs and microcontroller outputs.

The meaning assigned to each LED should be clearly defined in the program and documented in the final report. The indications should correspond to the actual conditions that have been tested.

4.2.9 BREADBOARD

A breadboard is used to establish temporary electrical connections between the ESP32, sensors, and other components without soldering.

It helps during prototype development because circuit connections can be changed easily while testing and debugging the system.

The power and ground connections must be arranged carefully to ensure that the sensors receive the correct supply voltage. Loose connections and incorrect wiring may cause unstable sensor readings.

For a final demonstration, the connections should be secured properly to prevent accidental disconnection during operation.

4.2.10 JUMPER WIRES

Jumper wires are used to connect the ESP32 to the sensors, HX711 modules, LEDs, and breadboard.

Different connections are required for power, ground, data, clock, and signal lines. The wire arrangement depends on the pin assignments and the

specifications of each component.

Proper wiring is important for reliable operation. The connections should be checked before powering the circuit to prevent short circuits or damage to the components.

4.3 SUMMARY

The hardware and software components work together to support the proposed waste analysis system. The gas, moisture, and metal sensors provide information about waste characteristics, while the load cells and HX711 modules measure weight. The level sensors monitor the filling condition of the bins.

The ESP32 processes the sensor inputs and controls the LEDs according to the implemented program. By combining these measurements, the system aims to identify possible incorrect waste disposal and provide information about the condition of both bins.

The performance of the system depends on sensor calibration, correct circuit connections, suitable detection rules, and testing with different waste samples.

CHAPTER 5

PROPOSED SYSTEM

5.1 OVERALL SYSTEM ARCHITECTURE

The proposed Intelligent Solid Waste Segregation Analysis system is designed to analyze waste placed in biodegradable and non-biodegradable bins. Its main purpose is to identify possible incorrect waste disposal by examining sensor readings and monitoring the condition of each bin.

The system consists of two waste bins, an ESP32 microcontroller, a gas sensor, two moisture sensors, two metal sensors, two load cells, two HX711 modules, two level sensors, and LEDs for indication. The sensors collect information about waste characteristics, weight, and filling level.

The moisture sensors provide information about the moisture content of the waste, while the metal sensors detect metallic materials. The gas sensor detects certain gases present near the waste. The load cells measure the weight of the waste through the HX711 modules, and the level sensors monitor how full the bins are.

The ESP32 receives the sensor readings and processes them according to the programmed detection rules. When the readings indicate a possible mismatch between the waste and the bin category, the system can provide a visual indication through the LEDs.

The proposed architecture combines sensor-based waste analysis with bin monitoring. It is intended to reduce the need for frequent manual inspection and help users identify possible disposal errors.

Main Components of the Architecture

1. **Biodegradable waste bin:** Intended for food scraps, vegetable waste, and other biodegradable materials.
2. **Non-biodegradable waste bin:** Intended for materials such as plastic, glass, and metal, according to the waste categories adopted in the project.

3. **Sensor unit:** Collects information about moisture, metal presence, and gases.
4. **Weight measurement unit:** Uses load cells and HX711 modules to measure the weight of waste in the bins.
5. **Bin-level monitoring unit:** Uses level sensors to monitor the filling condition of the bins.
6. **Processing unit:** Uses the ESP32 to process sensor readings and apply detection rules.
7. **Indication unit:** Uses LEDs to display the programmed system status.

The system does not automatically move waste between bins. Instead, it analyzes the sensor readings and provides indications of possible incorrect disposal.

5.2 WORKING PRINCIPLE AND DATA FLOW

The system operates by collecting sensor readings from the two waste bins and processing them using the ESP32 microcontroller. The readings are evaluated according to the conditions defined in the program.

Step 1: Waste Disposal

Users dispose of waste in either the biodegradable or non-biodegradable bin. The system monitors the waste present in each bin using the installed sensors.

Step 2: Sensor Data Collection

The moisture sensors measure the moisture characteristics of the waste. The metal sensors detect metallic materials, and the gas sensor detects certain gases near the waste.

The load cells measure the weight applied to them. The HX711 modules convert the load-cell signals into digital readings that can be processed by the ESP32. The level sensors provide information about the filling condition of the bins.

Step 3: Data Processing

The ESP32 reads the available sensor outputs and evaluates them using the programmed conditions. The sensor readings are used to determine whether the detected characteristics are consistent with the expected bin category.

For example, the detection of metal in the biodegradable bin may indicate possible incorrect disposal. Similarly, moisture measurements can help identify waste characteristics that differ from the expected conditions.

However, moisture and gas readings alone cannot conclusively identify all biodegradable materials. The system therefore uses the defined sensor conditions to indicate possible mismatches rather than claiming perfect identification of every waste item.

Step 4: Waste Mismatch Identification

The ESP32 compares the sensor readings with the detection rules implemented in the program. If the readings satisfy a condition associated with possible incorrect disposal, the system generates the corresponding indication.

The detection rules must be tested using different waste samples. Suitable thresholds and combinations of sensor readings should be selected based on the actual performance of the components.

Step 5: LED Indication

The LEDs provide visual indications according to the program. They may be configured to indicate a possible waste mismatch or a high bin-filling level.

The exact meaning of each LED depends on the implemented circuit and program. The final report should describe the actual indication logic used in the prototype.

Step 6: Weight and Level Monitoring

The load cells provide weight measurements, while the level sensors monitor the filling condition of the bins. These measurements help users understand the amount of waste collected and determine when a bin may need to be emptied.

Data Flow

The overall data flow can be represented as follows:

Waste deposited in the bins → Sensor readings collected → ESP32 processes the readings → Detection rules evaluated → LEDs provide indications → Bin weight and filling condition monitored.

This process allows the system to analyze waste characteristics and monitor both bins through a single microcontroller-based setup.

5.3 HARDWARE SUBSYSTEM OVERVIEW

The hardware subsystem consists of the sensors, microcontroller, weight measurement units, indication devices, and electrical connections.

5.3.1 Sensor Subsystem

The sensor subsystem collects information about the waste in the two bins. The moisture and metal sensors support the analysis of waste characteristics. The gas sensor provides information about gases near the waste.

The two level sensors monitor the filling condition of the bins, while the two load cells measure the weight of the collected waste.

Each sensor must be installed in a suitable position to obtain useful readings. Sensor placement, environmental conditions, and the characteristics of the waste may influence the measurements.

5.3.2 Processing Subsystem

The ESP32 acts as the main controller of the proposed system. It receives sensor outputs and executes the program that interprets the readings.

The controller can evaluate digital signals, process supported analog inputs, obtain weight readings from the HX711 modules, and control the LEDs.

The ESP32 must be configured with suitable input and output pins. Voltage compatibility and the current requirements of the connected components must be considered during circuit design.

5.3.3 Weight Measurement Subsystem

The weight measurement subsystem consists of two load cells and two HX711 modules. Each load cell is connected to an HX711 module, which provides digital weight-related readings to the ESP32.

The load cells must be calibrated to obtain weight values in suitable units. The measurements can then be used to monitor the waste quantity in each bin.

5.3.4 Indication Subsystem

The indication subsystem consists of LEDs controlled by the ESP32. The LEDs provide information about the conditions detected by the system.

The program determines when the LEDs should be switched on or off. The indications may be associated with possible waste mismatch or bin-level warnings, depending on the implemented logic.

5.3.5 Electrical Connection Subsystem

The breadboard and jumper wires provide the connections between the ESP32 and the other hardware components.

The circuit includes power, ground, sensor signal, and communication connections where required. Correct connections are essential for stable sensor operation and reliable measurements.

The circuit should be checked carefully before the system is powered on. Any sensor requiring a voltage higher than the ESP32 input limits must be interfaced using an appropriate circuit.

5.4 SOFTWARE SUBSYSTEM OVERVIEW

The software subsystem controls the operation of the hardware and processes the sensor readings. The ESP32 program defines how the inputs are collected, evaluated, and converted into output indications.

5.4.1 Sensor Reading

The program reads data from the connected sensors. The method used depends on the type of output provided by each sensor. Digital sensors provide

digital states, analog sensors provide voltage-related readings, and the HX711 modules provide digital measurements from the load cells.

The sensor readings are collected at suitable intervals to support continuous monitoring.

5.4.2 Data Processing

After collecting the readings, the ESP32 processes the available values. The program may apply thresholds, compare sensor states, and combine multiple readings to identify possible incorrect disposal.

For example, a metal detection signal from the biodegradable bin may trigger a warning according to the programmed rules. Other sensor readings can provide additional information about the waste.

5.4.3 Detection Logic

The detection logic defines the conditions under which the system identifies a possible waste mismatch. These conditions must be based on actual sensor capabilities and testing.

The gas, moisture, and metal sensor readings should not be treated as conclusive proof of the complete waste composition. A reliable implementation should account for possible false detections and variations in sensor readings.

5.4.4 Output Control

The program controls the LEDs according to the detection results. It can also process the level sensor outputs to indicate when a bin approaches its selected filling limit.

The output conditions should be clearly defined so that each indication has a specific meaning.

5.4.5 System Monitoring

During development, sensor values can be displayed through the serial monitor if serial output is implemented in the program. This helps with checking the sensor connections, calibrating the load cells, and selecting suitable detection thresholds.

The system can be tested with different waste samples to evaluate whether the sensor readings and LED indications correspond to the expected conditions.

5.5 SUMMARY

The proposed system combines multiple sensors with an ESP32 microcontroller to analyze waste characteristics and monitor biodegradable and non-biodegradable bins. The gas, moisture, and metal sensors provide information about the waste, while the load cells and level sensors monitor weight and filling conditions.

The ESP32 processes the sensor readings and controls the LEDs according to the implemented detection rules. The system aims to identify possible incorrect waste disposal and provide useful information about bin conditions.

The proposed approach provides a foundation for developing a low-cost waste monitoring system. Its effectiveness depends on appropriate sensor placement, calibration, detection logic, and practical testing with different waste materials.

CHAPTER 6

HARDWARE DESIGN AND IMPLEMENTATION

6.1 COMPONENT SELECTION CRITERIA

The hardware components for the Intelligent Solid Waste Segregation Analysis system are selected based on their function, availability, cost, compatibility, and suitability for a small-scale prototype. The system requires sensors to analyze waste characteristics, components to measure weight and bin level, and a microcontroller to process the collected data.

The ESP32 is used as the main controller because it supports sensor interfacing and provides sufficient processing capability for the programmed detection logic. Different sensors are used to obtain information about moisture, metal presence, gases, waste weight, and bin filling level.

The selection of components also considers the ease of circuit assembly and testing. A breadboard and jumper wires allow the components to be connected and modified during development.

6.1.1 ESP32 Microcontroller Selection

The ESP32 is selected as the main processing unit because it supports digital input/output operations and analog measurements on suitable pins. It can receive sensor readings, execute the programmed conditions, and control the LEDs.

Its built-in Wi-Fi and Bluetooth capabilities also allow future expansion to wireless monitoring. However, wireless communication is not assumed to be implemented in the current prototype unless it is included in the actual program.

6.1.2 Sensor Selection

The gas sensor is included to detect certain gases near the waste. The moisture sensors measure moisture-related characteristics, while the metal sensors detect metallic materials.

These sensors provide different types of information that can be considered when identifying possible incorrect waste disposal. Using multiple sensors allows

the system to examine more than one characteristic of the waste.

The sensor models must be selected according to the intended measurement, operating voltage, output type, and compatibility with the ESP32.

6.1.3 Load Cell and HX711 Selection

Load cells are used to measure the weight of the waste in each bin. Since the electrical signals from load cells are small, HX711 modules are used to amplify and convert these signals into digital measurements.

Using a separate load cell and HX711 module for each bin allows the weight of the two bins to be monitored independently.

6.1.4 Level Sensor Selection

Level sensors are used to monitor how much space remains inside each bin. They provide information about the filling condition and can help indicate when a bin approaches its capacity.

The exact interfacing method depends on the sensor type used in the prototype.

6.1.5 Selection of Indication and Connection Components

LEDs are selected to provide visible output indications. A breadboard and jumper wires are used to connect the electronic components during assembly and testing.

These components support the construction of a simple prototype without requiring a permanently soldered circuit during the initial development stage.

6.2 SENSOR AND LOAD CELL INTERFACING

Sensor interfacing involves connecting the sensors to the ESP32 so that their readings can be collected and processed. Each component must be connected according to its electrical specifications and output type.

6.2.1 Gas Sensor Interfacing

The gas sensor is connected to the ESP32 using the signal connection provided by the sensor module. Depending on the model, the sensor may provide

an analog output, a digital output, or both.

If an analog output is used, it must be connected to a suitable ESP32 analog input and remain within the permitted input voltage range. If a digital output is used, the program reads its HIGH or LOW state.

The program evaluates the readings according to the selected detection conditions. The sensor should be allowed to stabilize according to its specifications before its readings are used.

6.2.2 Moisture Sensor Interfacing

The two moisture sensors are connected so that moisture-related readings can be obtained for the two-bin arrangement.

The output from each sensor is read independently by the ESP32. The program can compare the readings with predefined thresholds to determine whether the measured moisture characteristics fall within the selected ranges.

The thresholds should be established through practical testing. The relationship between output voltage and moisture content depends on the sensor model and its operating conditions.

6.2.3 Metal Sensor Interfacing

The two metal sensors are used to detect metallic materials in the bins. Their output signals are connected to suitable ESP32 input pins.

When a metal sensor detects a metallic object within its sensing range, it produces an output according to its design. The ESP32 reads this output and evaluates it using the programmed conditions.

The operating voltage and output type must be checked before connecting the sensors. If a sensor's output voltage exceeds the ESP32 input limit, suitable signal conditioning is required.

6.2.4 Load Cell and HX711 Interfacing

Each load cell is connected to an HX711 module. The module receives the electrical signal from the load cell and converts it into digital readings.

The HX711 is then connected to the ESP32 using its data and clock signal

connections. The program reads the module output and converts the readings into weight values using a calibration factor.

The load cells should be mounted securely so that the weight is transferred correctly. The system should be calibrated using known weights before measurements are recorded.

6.2.5 Level Sensor Interfacing

The two level sensors are used to monitor the filling condition of the two bins. Each sensor is connected to the appropriate ESP32 input according to its output type.

The program reads the sensor output and determines whether the measured level has reached a predefined limit. An LED indication can be generated if the programmed full-bin condition is satisfied.

The sensor placement and detection range should be tested using the actual bin structure.

6.2.6 LED Interfacing

The LEDs are connected to ESP32 output pins using appropriate current-limiting resistors where required.

The program controls the LED states based on the sensor readings and programmed conditions. The meaning of each indication should be defined clearly to distinguish possible waste mismatch from a bin-level warning.

6.3 MICROCONTROLLER PROGRAMMING

The ESP32 program controls the sensor reading, data processing, and LED indication functions. The program is developed according to the output type of each sensor and the detection rules required by the project.

6.3.1 Program Initialization

During initialization, the program configures the sensor input pins, LED output pins, and communication connections required by the HX711 modules.

The serial communication interface may also be initialized to display

sensor readings during testing.

6.3.2 Sensor Data Acquisition

After initialization, the program reads the connected sensors. Digital sensors are read as digital states, while analog sensors are read using suitable analog input functions.

The HX711 modules provide the load-cell measurements. These readings are processed separately to obtain weight estimates for the two bins.

6.3.3 Data Evaluation

The collected readings are evaluated according to the conditions defined in the program. For example, a metal detection signal in the biodegradable bin may be treated as a possible waste mismatch.

Moisture and gas readings can provide additional information, but they should not be treated as definitive identification of every material. The detection rules must be validated through experiments.

6.3.4 Output Control

The ESP32 controls the LEDs according to the results of the evaluation. If the sensor readings satisfy a programmed warning condition, the corresponding LED is activated.

The level sensor outputs can also be used to indicate when the bin reaches the selected filling limit.

6.3.5 Calibration and Testing

Calibration is required to improve the reliability of the sensor readings. The load cells are calibrated using known weights, while moisture and level sensors are tested under different conditions.

Metal detection is checked using suitable metallic samples. Gas sensor readings are observed under appropriate test conditions.

The results are used to verify whether the program responds correctly to the sensor inputs.

6.4 CIRCUIT DESIGN AND POWER SUPPLY

The circuit design connects the sensors, HX711 modules, ESP32, and LEDs into a working prototype. The breadboard and jumper wires are used to establish the connections during assembly.

6.4.1 Circuit Design Considerations

The circuit should provide the correct power and ground connections to all components. Sensor signal connections must be assigned to suitable ESP32 pins according to their input and output requirements.

The circuit design should consider the following points:

1. Compatibility between the component supply voltage and the ESP32.
2. Correct connection of signal and ground wires.
3. Suitable input pins for analog and digital sensors.
4. Separate data and clock connections for the HX711 modules.
5. Proper LED connection and current limiting.
6. Secure mounting of load cells.
7. Avoidance of loose connections and short circuits.

6.4.2 Power Supply

The ESP32 and sensors require a suitable power supply. The supply method must be selected according to the specifications of the development board and connected modules.

The operating voltage of each sensor should be verified before the circuit is powered. The ESP32 uses 3.3 V logic, and its input pins must not be exposed to voltages beyond their permitted limits.

If a sensor requires a different supply voltage, the circuit must provide the appropriate supply and compatible signal levels.

6.4.3 Breadboard Assembly

The breadboard is used to connect the components during prototype development. Power and ground connections are distributed to the required components, while jumper wires connect the signal pins.

Before powering the system, each connection should be checked against the circuit diagram. This helps prevent wiring errors and possible damage to the hardware.

6.4.4 Load Cell Mounting

The load cells must be mounted according to their mechanical design. The bin's weight should be transferred through the intended sensing point without unwanted contact that could affect the readings.

After mounting, the load cells should be checked with known weights. Calibration should be performed again if the mechanical arrangement changes significantly.

6.5 COMMUNICATION SETUP

The ESP32 communicates with the connected components through their supported electrical interfaces. The communication method depends on the type of sensor and module.

The HX711 modules use data and clock signals to transfer load-cell measurements to the ESP32. Other sensors may use analog outputs, digital signals, or their own communication interfaces.

During testing, serial communication can be used to display sensor readings on a computer, provided that this feature is implemented in the program. This supports debugging and calibration.

The ESP32 also supports Wi-Fi and Bluetooth, but these features are optional for the current project. A cloud platform, mobile application, MQTT connection, or remote dashboard should only be described as implemented if it is actually developed and tested.

6.6 HARDWARE ASSEMBLY PROCEDURE

The hardware assembly follows a sequence to simplify construction and testing.

1. Place the ESP32 development board on the working platform.
2. Connect the power and ground lines to the breadboard.
3. Connect the gas sensor and verify its output.
4. Connect the two moisture sensors and test their readings.
5. Connect the two metal sensors and verify their detection outputs.
6. Connect each load cell to its corresponding HX711 module.
7. Connect the HX711 modules to the ESP32 and calibrate the weight readings.
8. Install and connect the two level sensors.
9. Connect the LEDs to the selected output pins.
10. Upload the program to the ESP32.
11. Test each sensor individually before testing the complete system.
12. Verify the LED indications under the programmed conditions.

The assembly should be performed with the power disconnected whenever wiring changes are made.

6.7 SUMMARY

This chapter describes the selection, interfacing, and implementation of the hardware used in the Intelligent Solid Waste Segregation Analysis system. The ESP32 acts as the controller, while the sensors provide information about waste characteristics, weight, and bin level.

The load cells and HX711 modules support weight measurement, and the LEDs provide visual indications based on the programmed logic. Proper circuit connections, calibration, and individual component testing are necessary for reliable operation.

The completed hardware forms the basis for integrating the sensors and evaluating possible incorrect waste disposal in the two-bin arrangement.

CHAPTER 7
SENSOR DATA ANALYSIS AND WASTE MISMATCH DETECTION LOGIC

7.1 INTRODUCTION

Sensor data analysis is an important part of the Intelligent Solid Waste Segregation Analysis system. The system collects information about the waste present in biodegradable and non-biodegradable bins using different sensors. The ESP32 processes these readings to identify possible incorrect waste disposal.

The project uses a gas sensor, two moisture sensors, two metal sensors, two load cells with HX711 modules, and two level sensors. Each component provides information about a particular characteristic of the waste or the condition of the bin.

The sensor readings are evaluated using programmed conditions. The metal sensors help detect metallic materials, the moisture sensors measure moisture-related characteristics, and the gas sensor detects certain gases near the waste. The load cells measure waste weight, while the level sensors monitor the filling condition of the bins.

The purpose of the analysis is to combine the available measurements and provide useful indications when the detected characteristics suggest a possible mismatch between the waste and its intended bin category.

7.2 SENSOR DATA COLLECTION

Sensor data collection is the process of obtaining readings from the connected hardware components. The ESP32 acts as the central controller and collects the available sensor outputs.

7.2.1 Gas Sensor Data

The gas sensor provides readings related to the gases present near the waste. These readings may change depending on the type of gas, its

concentration, ventilation, and environmental conditions.

The ESP32 reads the sensor output and evaluates it according to the conditions defined in the program. Gas readings can provide additional information about the waste environment but cannot independently confirm whether a material is biodegradable.

7.2.2 Moisture Sensor Data

The two moisture sensors provide moisture-related readings for the two-bin arrangement. These readings can help distinguish between waste with different moisture characteristics.

The measured values may vary depending on the waste material, water content, sensor placement, and contact conditions. Therefore, suitable thresholds should be established through practical testing.

7.2.3 Metal Sensor Data

The two metal sensors provide signals when metallic materials are detected within their operating range.

The ESP32 reads these signals and checks whether metal has been detected in either bin. For example, metal detection in the biodegradable bin may be treated as a possible indication of incorrect disposal.

The detection result depends on the sensor type, sensing distance, material, and installation position.

7.2.4 Weight Data

The load cells measure the weight applied to them. Their electrical outputs are processed by the HX711 modules and converted into digital readings.

The ESP32 uses these readings to estimate the weight of the waste in each bin. Calibration is necessary to convert the raw measurements into meaningful weight values.

Weight measurements help monitor waste quantity, but weight alone cannot determine the material composition of the waste.

7.2.5 Bin-Level Data

The two level sensors provide information about the filling condition of the bins. The ESP32 reads the sensor outputs and checks whether the configured bin-level condition has been reached.

This information can be used to provide an indication when a bin approaches its capacity. The exact processing method depends on the level sensor model.

7.3 SENSOR CALIBRATION AND PREPROCESSING

Calibration and preprocessing help improve the usefulness of the collected sensor data. Raw readings may vary because of sensor characteristics, environmental conditions, electrical noise, or installation differences.

7.3.1 Moisture Sensor Calibration

The moisture sensors should be tested using samples with different moisture characteristics. The readings can be recorded and compared to determine suitable thresholds for the intended application.

Calibration should be performed under conditions similar to the actual waste-bin setup. The selected threshold values should be documented after testing.

7.3.2 Metal Sensor Verification

The metal sensors should be tested using suitable metallic samples placed at different positions within the sensing range.

The purpose is to verify whether the sensor output changes correctly when metal is present. Testing with non-metallic samples can help identify unwanted detections.

7.3.3 Load Cell Calibration

Load cell calibration involves comparing the sensor readings with known weights.

The basic procedure is:

1. Place the empty bin or measurement platform in its intended position.
2. Record the unloaded reading.
3. Place a known weight on the platform.
4. Record the corresponding sensor reading.
5. Adjust the calibration factor in the program.
6. Repeat the test with additional known weights.

The calibration process should be repeated if the load cell mounting or mechanical arrangement changes.

7.3.4 Gas Sensor Stabilization

The gas sensor should be operated according to its manufacturer's instructions before its readings are used. Some gas sensors require a warm-up or stabilization period.

Readings should be observed under different test conditions to understand the sensor's normal output and response to the gases it detects.

7.3.5 Level Sensor Verification

The level sensors should be tested at different filling conditions. Their outputs should be compared with the actual waste level or distance measured by the sensor.

The selected threshold should correspond to the intended warning level of the bin.

7.4 WASTE MISMATCH DETECTION LOGIC

Waste mismatch detection is the process of identifying sensor readings that may indicate that the contents of a bin do not match its intended category.

The ESP32 evaluates the readings using predefined conditions. These conditions should be developed and tested according to the actual capabilities of the installed sensors.

7.4.1 Biodegradable Bin Analysis

The biodegradable bin is intended for materials such as food scraps, vegetable waste, and other biodegradable waste.

If the metal sensor detects a metallic object in this bin, the program may identify the condition as a possible disposal error. Moisture and gas readings can provide additional information about the waste characteristics.

However, a high or low moisture reading does not conclusively establish whether a material is biodegradable. Similarly, the absence of metal does not prove that the waste is biodegradable.

7.4.2 Non-Biodegradable Bin Analysis

The non-biodegradable bin is intended for materials such as plastic, glass, and metal, according to the categories used in the project.

If the sensors detect characteristics associated with unwanted biodegradable waste, the ESP32 can evaluate those readings according to the programmed rules. Moisture and gas measurements may be considered as supporting indicators.

These readings should be interpreted carefully because some non-biodegradable materials may be wet or contaminated with food waste. The system should therefore indicate a possible mismatch rather than claim definitive identification from a single sensor.

7.4.3 Combined Sensor Analysis

Combining the sensor readings can provide more information than considering a single measurement.

For example, the system can evaluate metal detection along with moisture readings and the other available sensor outputs. The program then checks the combined information against the defined conditions.

The load cells and level sensors primarily provide information about bin weight and filling condition. These measurements support bin monitoring but do not independently determine whether the waste is biodegradable.

The accuracy of combined sensor analysis depends on the detection rules, sensor performance, calibration, and testing.

7.5 LED INDICATION LOGIC

The LEDs provide visual information about the conditions detected by the system. Their operation is controlled by the ESP32 program.

An LED may be configured to indicate a possible waste mismatch, while another indication may be used for a bin-level warning. The actual functions depend on the circuit and the program implemented in the prototype.

The indication logic should be designed so that each LED state has a clearly defined meaning. Sensor readings should be checked before an indication is generated to avoid unnecessary warnings.

During testing, the LED states should be compared with the sensor outputs and the actual test conditions. Any differences should be recorded and investigated.

7.6 ALGORITHM FOR SENSOR DATA ANALYSIS

The following general algorithm describes the intended operation of the system.

1. Start the ESP32.
2. Initialize the sensor connections and LED output pins.
3. Read the sensor outputs for both bins.
4. Obtain weight measurements from the load cells through the HX711 modules.
5. Read the bin-level sensor outputs.
6. Evaluate the moisture and metal sensor readings.
7. Evaluate the gas sensor reading according to the implemented conditions.
8. Compare the available readings with the programmed detection rules.
9. Identify any condition that suggests possible incorrect disposal.

10. Activate the appropriate LED indication.
11. Check the weight and filling condition of each bin.
12. Repeat the process at the next reading interval.

The algorithm describes the overall processing sequence. The exact decision conditions and thresholds must match the program used in the working prototype.

7.7 LIMITATIONS OF SENSOR DATA ANALYSIS

The performance of the system depends on the type and quality of the sensors and the way they are installed.

Moisture sensors measure moisture-related properties but cannot distinguish every biodegradable material from non-biodegradable material. Metal sensors detect metals within their operating range but cannot identify all other waste types. Gas sensors respond to particular gases and environmental conditions rather than directly identifying waste categories.

Load cells measure weight, while level sensors monitor bin filling conditions. These measurements are useful for monitoring but do not independently establish the composition of waste.

Mixed waste, contaminated materials, sensor noise, and changes in environmental conditions may affect the results. Therefore, the system should be tested with different waste samples, and its detection rules should be adjusted according to the observed performance.

The system is intended to provide indications of possible incorrect disposal rather than guarantee perfect classification of every waste item.

7.8 SUMMARY

This chapter describes the collection and analysis of sensor readings in the proposed waste monitoring system. The ESP32 processes the gas, moisture, metal, weight, and level measurements according to the programmed conditions.

The combined sensor readings help identify possible waste disposal errors and monitor the condition of the two bins. Calibration and testing are necessary to establish suitable detection thresholds and improve reliability.

The analysis forms an important part of the project because it connects the physical sensors with the decision logic used to provide visual indications.

CHAPTER 8
SYSTEM INTEGRATION AND MONITORING

8.1 Introduction

System integration is the process of connecting the hardware components and software logic so that they work together as a complete system. In the Intelligent Solid Waste Segregation Analysis project, the ESP32 acts as the main controller. It receives readings from the connected sensors and processes the information to analyze the waste present in the biodegradable and non-biodegradable bins.

The system combines gas sensors, moisture sensors, metal sensors, load cells, HX711 modules, level sensors, and LEDs. Each component performs a specific function in monitoring the waste. The combined sensor readings help the system identify possible incorrect disposal and indicate the status of the bins.

The main purpose of integration is to ensure that the sensors provide useful readings, the controller processes the readings correctly, and the LEDs provide suitable visual indications.

8.2 Hardware Integration

Hardware integration involves connecting the different electronic components to the ESP32 according to the circuit design.

The two bins are used to collect biodegradable and non-biodegradable waste separately. The sensors are arranged according to their intended functions. Moisture sensors help measure moisture-related characteristics, metal sensors detect metallic objects, gas sensors detect gases near the waste, load cells measure weight, and level sensors monitor the filling level of the bins.

The HX711 modules are used to interface the load cells with the ESP32. The ESP32 processes the sensor readings and controls the LEDs according to the programmed logic.

8.3 Sensor and Controller Integration

The ESP32 is the central processing unit of the proposed system. It receives readings from the connected sensors and uses the programmed conditions to interpret them.

The moisture sensor readings provide information about the moisture present in the waste. The metal sensors help detect metallic materials. The gas sensor provides information about the presence of gases near the waste. The load cells measure the weight of the waste, while the level sensors indicate how full the bins are.

The controller combines the available readings to check whether the observed waste characteristics may indicate incorrect disposal. For example, metal detection in a biodegradable waste bin may indicate that a metallic object has been placed in the wrong bin.

However, individual sensor readings cannot identify every type of waste accurately. Therefore, the system should indicate a possible mismatch rather than treating every sensor reading as a definite identification of waste type.

8.4 Bin Monitoring

Bin monitoring helps observe the condition of the biodegradable and non-biodegradable waste bins. The system uses the connected sensors to collect information about the waste and the filling level.

The load cells provide weight measurements that help monitor the amount of waste collected. The level sensors provide information about the available space inside the bins. These readings can help determine when a bin is becoming full.

The moisture, metal, and gas sensor readings provide additional information about the characteristics of the collected waste. The ESP32 evaluates these readings using the programmed conditions.

Regular monitoring can help users identify possible disposal mistakes and recognize when the bins require attention. The system supports waste monitoring but does not physically transfer waste from one bin to another.

8.5 Alert Generation Using LEDs

LEDs are used to provide visual indications based on the programmed sensor conditions. They help users understand the system's status without requiring a complex display.

When the sensor readings satisfy a condition associated with possible incorrect disposal, the ESP32 can activate the corresponding LED indication. Different indications may also be programmed for normal operation or a bin that has reached its specified filling level.

The exact LED colours, activation conditions, and meanings depend on the implemented circuit and program. These details should be documented according to the actual prototype.

LED-based indications make the system simple to observe and suitable for demonstration in places such as colleges, offices, homes, and other waste collection areas.

8.6 Complete System Workflow

The complete system follows a sequence of sensor measurement, data processing, and visual indication.

1. Waste is placed in either the biodegradable or non-biodegradable bin.
2. The connected sensors measure the relevant characteristics of the waste.
3. The load cells measure the weight through the HX711 modules.
4. The level sensors monitor the filling condition of the bins.
5. The ESP32 receives the available sensor readings.
6. The programmed logic evaluates the readings according to predefined conditions.

7. The system identifies conditions that may indicate incorrect disposal.
8. The ESP32 activates the appropriate LED indication when the programmed conditions are satisfied.
9. The user observes the indication and can check the bin when necessary.

This workflow provides a basic method for monitoring waste characteristics and indicating possible segregation errors. The reliability of the result depends on sensor calibration, the placement of the sensors, the programmed thresholds, and the types of waste being tested.

8.7 Importance of System Integration

System integration is important because the individual components must operate together to achieve the project objectives. Incorrect wiring, unstable sensor readings, unsuitable thresholds, or programming errors may affect the overall output.

The integrated prototype provides a practical platform for observing sensor readings, testing the waste mismatch detection logic, and evaluating the visual indications. It also helps identify areas that require improvement during testing.

The system can be further improved by adding a display, storing sensor readings, or introducing wireless monitoring if these features are implemented in a future version.

8.8 Summary

This chapter described the integration of the sensors, ESP32 controller, load cells, HX711 modules, level sensors, and LEDs. The system collects sensor readings, processes the information using programmed conditions, monitors the bins, and provides visual indications of possible incorrect waste disposal. The integrated system offers a simple approach to waste monitoring while recognizing that sensor-based identification has limitations.

CHAPTER 9
RESULTS AND DISCUSSION

9.1 Introduction

The results and discussion chapter describes the testing and evaluation of the Intelligent Solid Waste Segregation Analysis system. The main purpose of testing is to verify whether the sensors collect useful readings, the ESP32 processes the readings correctly, and the LEDs provide the expected visual indications.

The system is designed to monitor biodegradable and non-biodegradable waste bins using gas sensors, moisture sensors, metal sensors, load cells, HX711 modules, and level sensors. The results obtained from these components can be used to evaluate the working of the prototype.

The testing process includes individual sensor testing, load measurement, bin-level monitoring, waste mismatch indication, and overall system integration. The observed results should be recorded during actual hardware testing.

9.2 Sensor Testing

Sensor testing is performed to verify that each sensor responds to the condition it is designed to measure or detect. Each sensor should be tested separately before evaluating the complete system.

9.2.1 Gas Sensor Testing

The gas sensor is tested to check whether its output changes when the gas conditions near the waste change. The sensor reading depends on the gas present, the sensor type, the surrounding environment, and its calibration.

The observed readings should be recorded under different test conditions. A gas sensor reading alone cannot confirm whether the waste is biodegradable or non-biodegradable.

9.2.2 Moisture Sensor Testing

The moisture sensors are tested using materials with different moisture conditions. The readings are observed to determine whether the sensors respond to changes in moisture.

The readings can help distinguish between materials with different moisture characteristics. However, wet non-biodegradable materials and dry biodegradable materials may produce misleading results. Therefore, moisture readings must be interpreted along with other available sensor readings.

9.2.3 Metal Sensor Testing

The metal sensors are tested using suitable metallic and non-metallic objects. The sensor output is observed to determine whether metallic objects can be detected within the sensor's effective detection range.

Metal detection can help identify possible metallic waste in a bin. However, it cannot identify all non-biodegradable materials, such as plastic, paper packaging, or glass.

9.2.4 Load Cell Testing

The load cells are tested by placing known weights on the weighing platform. The measured values are compared with the known weights to evaluate the measurement difference.

The HX711 modules interface the load cells with the ESP32. Proper calibration is necessary to obtain meaningful weight measurements.

9.2.5 Level Sensor Testing

The level sensors are tested by changing the amount or height of material inside the bins. Their output is observed to determine whether they respond to changes in the filling level.

These readings can help identify when a bin approaches its configured filling limit.

9.3 Test Procedure

The following procedure can be used to evaluate the prototype:

1. Check the connections between the ESP32, sensors, HX711 modules, and LEDs.
2. Power the system using the intended power supply.
3. Verify that each sensor produces readings.
4. Test the moisture sensors using materials with different moisture conditions.
5. Test the metal sensors using metallic and non-metallic objects.
6. Compare load cell measurements with known weights.
7. Test the level sensors at different bin-filling levels.
8. Place selected waste materials in the two bins separately.
9. Observe the sensor readings and LED indications.
10. Record the expected response and the actual response for each test.

Testing should be conducted carefully, keeping the sensor placement and test conditions as consistent as possible.

9.4 Test Cases and Observations

The following tables provide a format for recording the actual test results. The observation fields must be completed after performing the tests.

Table 9.1: Individual Sensor Testing

Test ID	Component	Test Condition	Expected Observation	Actual Observation
T1	Gas sensor	Different gas conditions	Sensor output changes according to its response	To be recorded

T2	Moisture sensor	Different moisture conditions	Readings respond to moisture changes	To be recorded
T3	Metal sensor	Metallic object nearby	Metal detection response	To be recorded
T4	Load cell	Known weight applied	Weight reading changes with load	To be recorded
T5	Level sensor	Different filling levels	Output reflects the tested level	To be recorded
T6	LED	Programmed trigger condition	LED responds according to the code	To be recorded

Table 9.2: Waste Monitoring Tests

Test ID	**Bin**	**Test Material**	**Purpose**	**Actual Result**
W1	Biodegradable	Selected biodegradable waste	Observe sensor response	To be recorded
W2	Biodegradable	Metallic waste	Check possible metal-related mismatch	To be recorded
W3	Non-biodegradable	Selected non-biodegradable waste	Observe sensor response	To be recorded
W4	Non-biodegradable	Selected biodegradable	Check for a possible	To be recorded

		waste	mismatch	
W5	Either bin	Different waste quantities	Check weight measurement	To be recorded
W6	Either bin	Increasing filling level	Check level monitoring	To be recorded

The actual result may differ depending on the sensor characteristics, waste composition, moisture content, calibration, and the programmed detection conditions.

9.5 Discussion of Results

The system's performance should be evaluated by examining the readings from each sensor and the corresponding LED indications.

If the moisture sensors respond consistently to different moisture conditions, they can provide useful information about the waste. If the metal sensors detect the tested metallic objects, they can help identify possible metallic contamination. Similarly, calibrated load cells can provide weight measurements, while level sensors can support bin-filling monitoring.

The ESP32 combines these inputs according to the implemented program. The LED output indicates whether the programmed conditions have been met. Correct operation of each individual component is important for reliable system performance.

The results should also be examined for unexpected readings, unstable outputs, incorrect indications, and differences between repeated tests. Any such issues should be documented and investigated before drawing conclusions about the system's performance.

9.6 Limitations Observed During Testing

Sensor-based waste analysis has several limitations. Moisture sensors

measure moisture-related properties rather than biodegradability itself. Metal sensors detect only materials within their effective detection range. Gas sensor readings may vary with environmental conditions and the composition of the waste.

Load cells measure weight but cannot determine the material type from weight alone. Level sensors monitor filling conditions rather than identifying the contents of a bin.

Consequently, the system may not correctly identify every disposal error. Its performance depends on the combination of sensors, calibration, sensor placement, the detection logic, and the waste materials used during testing.

9.7 Overall System Evaluation

The overall evaluation should consider the following aspects:

- **Sensor response:** Whether each sensor responds to the tested conditions.
- **Measurement reliability:** Whether repeated measurements are reasonably consistent.
- **Weight measurement:** Whether load cell readings are close to known weights after calibration.
- **Bin-level monitoring:** Whether the level sensors respond to changes in filling level.
- **LED indication:** Whether the LEDs operate according to the programmed conditions.
- **Integrated operation:** Whether the sensors and ESP32 operate together as intended.
- **Mismatch indication:** Whether the system indicates the selected test cases according to its programmed logic.

These criteria provide a structured method for evaluating the prototype without assuming that it achieves perfect waste classification.

9.8 Summary

This chapter presented the testing procedure and evaluation criteria for the Intelligent Solid Waste Segregation Analysis system. It covered individual sensor testing, weight measurement, bin-level monitoring, LED indications, and integrated system operation.

The actual test observations must be recorded before reporting the final performance of the system. The results will help identify the strengths and limitations of the prototype and determine areas that require further improvement.

CHAPTER 10
CONCLUSION AND FUTURE ENHANCEMENTS

10.1 Conclusion

The project titled **"Intelligent Solid Waste Segregation Analysis"** focuses on monitoring biodegradable and non-biodegradable waste bins and identifying possible incorrect waste disposal using sensor-based analysis. The system uses an ESP32 microcontroller along with gas sensors, moisture sensors, metal sensors, load cells, HX711 modules, level sensors, and LEDs.

Each sensor performs a specific function in collecting information about the waste. Moisture sensors provide moisture-related readings, metal sensors detect metallic objects, and the gas sensor detects gases near the waste. Load cells measure the weight of the collected waste, while level sensors monitor the filling condition of the bins. The ESP32 processes the available readings according to the programmed conditions and controls the LED indications.

The proposed system provides a simple approach to monitoring waste characteristics and alerting users to conditions that may indicate incorrect disposal. It can help create awareness about proper waste segregation in places such as colleges, homes, offices, and public areas.

The project also demonstrates the practical use of embedded systems and sensor integration in waste management. However, the system's ability to identify incorrect disposal depends on sensor calibration, the selected detection conditions, and the types of waste tested. Since the sensors cannot identify every material accurately, the system should be considered a waste-monitoring and mismatch-indication prototype rather than a perfect waste classification system.

Overall, the project provides a foundation for developing affordable sensor-based waste monitoring solutions that encourage better waste segregation practices.

10.2 Advantages of the Proposed System

The major advantages of the proposed system are:

1. It supports monitoring of biodegradable and non-biodegradable waste bins.
2. It uses commonly available electronic components.
3. It provides information about waste characteristics through multiple sensors.
4. It measures the weight of collected waste using load cells.
5. It monitors bin-filling levels using level sensors.
6. It provides visual indications through LEDs.
7. It can help users identify possible waste disposal mistakes.
8. It provides practical experience in ESP32 programming and sensor integration.
9. It can be developed as a prototype for educational and small-scale waste management applications.

10.3 Limitations of the System

Although the system provides useful monitoring features, certain limitations must be considered.

1. Moisture sensors measure moisture-related properties and cannot independently determine whether waste is biodegradable.
2. Metal sensors can detect suitable metallic objects but cannot identify all non-biodegradable materials.
3. Gas sensor readings may vary depending on environmental conditions and the composition of the waste.
4. Load cells measure weight but cannot determine waste type from weight alone.
5. The accuracy of the system depends on sensor calibration, placement, and programmed conditions.
6. The system may produce incorrect indications when different materials

have similar sensor characteristics.

7. The prototype does not automatically separate or transfer waste between bins.

These limitations provide opportunities for further development and improvement.

10.4 Future Enhancements

The following enhancements can be considered in future versions of the project.

10.4.1 Improved Waste Identification

Additional sensors and improved detection logic can be explored to distinguish waste materials more effectively. Testing with a wider variety of waste samples can help identify the conditions under which the current sensors perform reliably.

10.4.2 Display Unit Integration

An LCD or OLED display can be added to show sensor readings, measured weight, bin-filling status, and warning messages. This can make the system easier for users to understand.

10.4.3 Wireless Monitoring

Wireless communication using the ESP32 can be explored to transmit sensor readings to a monitoring interface. This feature would allow users to view bin conditions remotely if implemented with suitable software and connectivity.

10.4.4 Data Storage and Analysis

A data storage feature can be introduced to record sensor readings, weight measurements, and bin-filling information over time. The collected data could be used to study waste generation patterns and evaluate system behaviour.

10.4.5 Improved Calibration and Testing

The system can be tested using known weights, different moisture conditions, metallic and non-metallic materials, and different bin-filling levels.

Calibration and repeated testing can help improve measurement consistency and reduce incorrect indications.

10.4.6 Automatic Waste Sorting

A mechanical sorting mechanism could be investigated in a future version. Suitable actuators and a sorting arrangement would be required to physically separate waste. This functionality is not part of the current sensor-monitoring prototype.

10.4.7 Power Efficiency

The system's power consumption can be studied, and suitable power-saving techniques can be considered. This may help support longer operation in waste collection areas.

10.5 Summary

This chapter presented the conclusion, advantages, limitations, and possible future enhancements of the Intelligent Solid Waste Segregation Analysis project. The system combines multiple sensors with an ESP32 controller to monitor waste characteristics, measure weight, check bin-filling levels, and provide LED indications.

The project demonstrates how sensor-based embedded systems can support waste monitoring and help identify possible disposal errors. Future improvements in sensor selection, calibration, data storage, wireless monitoring, and waste identification may increase its usefulness and reliability.

CHAPTER 11
REFERENCES

The following references provide technical information related to the microcontroller, load cell interface, programming environment, and solid waste management concepts used in this project.

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5. Central Pollution Control Board (CPCB), *Centralised Portal for Solid Waste Management*. Available: https://www.swm.cpcb.gov.in/
6. Ministry of Environment, Forest and Climate Change, Government of India, *Solid Waste Management Rules, 2026*. Information available through the Central Pollution Control Board portal: https://www.swm.cpcb.gov.in/

Reference Usage

The ESP32 datasheet provides technical information about the controller and its input/output interfaces. The HX711 datasheet describes the load cell interface and its weight-measurement functionality. Arduino documentation provides guidance on programming tools and microcontroller development. UNEP and CPCB resources provide background information on waste management and waste segregation.

The references should be cited in the relevant chapters wherever technical information or background facts are used. The exact software documentation should be updated if a different programming environment was used in the implementation.

CHAPTER 12
APPENDICES

APPENDIX A: HARDWARE COMPONENTS LIST

The following components are used in the Intelligent Solid Waste Segregation Analysis project.

S. No.	Component Name	Quantity	Purpose
1	Gas Sensor	1	Detects gases near the waste
2	Moisture Sensor	2	Measures moisture-related characteristics
3	Metal Sensor	2	Detects metallic objects
4	Load Cell	2	Measures the weight of the waste
5	HX711 Module	2	Interfaces load cells with the controller
6	ESP32	1	Processes sensor readings and controls outputs
7	Breadboard	1	Supports prototype circuit connections
8	Jumper Wires	As required	Connects the electronic components
9	LEDs	As required	Provides visual status indications
10	Level Sensor	2	Monitors the filling level of the bins

APPENDIX B: SYSTEM INPUTS AND OUTPUTS

The system receives information from the sensors and generates visual indications based on the programmed conditions.

S. No.	Input Component	Input Information	System Use
1	Gas Sensor	Gas-related sensor reading	Supports analysis of waste conditions
2	Moisture Sensors	Moisture-related readings	Supports analysis of moisture characteristics
3	Metal Sensors	Metal detection status	Helps identify possible metallic contamination
4	Load Cells and HX711 Modules	Weight measurements	Monitors collected waste weight
5	Level Sensors	Filling-level information	Monitors the condition of the bins

APPENDIX C: GENERAL SYSTEM ALGORITHM

Step 1: Start the system.

Step 2: Initialize the ESP32 and the connected components.

Step 3: Read the available sensor values.

Step 4: Obtain weight measurements through the HX711 modules.

Step 5: Read the bin-filling information from the level sensors.

Step 6: Check the sensor readings against the conditions defined in the program.

Step 7: Determine whether the readings indicate a possible mismatch or another monitored condition.

Step 8: Activate the appropriate LED according to the programmed logic.

Step 9: Continue monitoring the sensors.

Step 10: Repeat the process while the system is operating.

APPENDIX D: HARDWARE TESTING RECORD

Project Title: Intelligent Solid Waste Segregation Analysis

Date of Testing: _____

Tested By: _____

S. No.	Component Tested	Test Condition	Observed Reading or Response	Status
1	Gas Sensor	_____	_____	_____
2	Moisture Sensor 1	_____	_____	_____
3	Moisture Sensor 2	_____	_____	_____
4	Metal Sensor 1	_____	_____	_____
5	Metal Sensor 2	_____	_____	_____
6	Load Cell 1	_____	_____	_____
7	Load Cell 2	_____	_____	_____
8	Level Sensor 1	_____	_____	_____

9	Level Sensor 2	_____	_____	_____
10	LED Indications	_____	_____	_____

APPENDIX E: PHOTOGRAPHS OF THE PROJECT

Insert the actual photographs of the developed prototype in this section.

Figure E.1: Complete hardware prototype.

Figure E.2: ESP32 controller and circuit connections.

Figure E.3: Gas and moisture sensor arrangement.

Figure E.4: Metal sensor arrangement.

Figure E.5: Load cells and HX711 modules.

Figure E.6: Level sensors installed in the bins.

Figure E.7: LED indication during testing.

APPENDIX F: PROGRAM SOURCE CODE

Insert the final ESP32 program used in the project in this section.

The source code should include the following parts, where applicable:

1. Component and pin initialization.
2. Sensor reading functions.
3. Load cell reading and calibration logic.
4. Sensor condition checking.
5. Waste mismatch indication logic.
6. LED control logic.
7. Main monitoring loop.

The submitted source code should match the program used to operate the hardware. Include any required library names and explain the purpose of important functions if necessary.

APPENDIX H: PROJECT SUMMARY

The Intelligent Solid Waste Segregation Analysis project demonstrates the integration of sensors and an ESP32 microcontroller for monitoring biodegradable and non-biodegradable waste bins. The system uses sensor readings to analyze waste characteristics, measure weight, monitor bin-filling levels, and provide LED indications according to the programmed conditions.

The appendices provide supporting information about the hardware components, system inputs and outputs, general algorithm, testing records, prototype photographs, and source code. These materials support the documentation and evaluation of the completed project.

SDG MAPPING

SDG 12 – Responsible Consumption and Production: The project supports responsible waste disposal by monitoring biodegradable and non-biodegradable waste bins and identifying possible waste segregation errors.

SDG 11 – Sustainable Cities and Communities: The project supports better waste management practices through waste-bin monitoring and encourages proper waste disposal in colleges, homes, offices, and public places.

SDG 9 – Industry, Innovation and Infrastructure: The project applies sensor technology and the ESP32 microcontroller to develop a technology-based solution for waste monitoring.